

Tackling the Many-Body Problem: the DFT Approach

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Outline

- The many-electron problem (ground states)
- (Generalized-) Kohn-Sham DFT
- A bird's-eye view of density functional approximations

THE MANY-ELECTRON PROBLEM

(ground states)

State of electrons = solution of Schrödinger equation

$$\hat{H}\Psi_E = E\Psi_E$$

Electronic wavefunction (BOA)

$$\Psi_{\underline{\mathbf{R}}}(\underline{\mathbf{x}})$$

$$\underline{\mathbf{R}} = \{\mathbf{R}_A\}_{A=1}^{M=3}$$

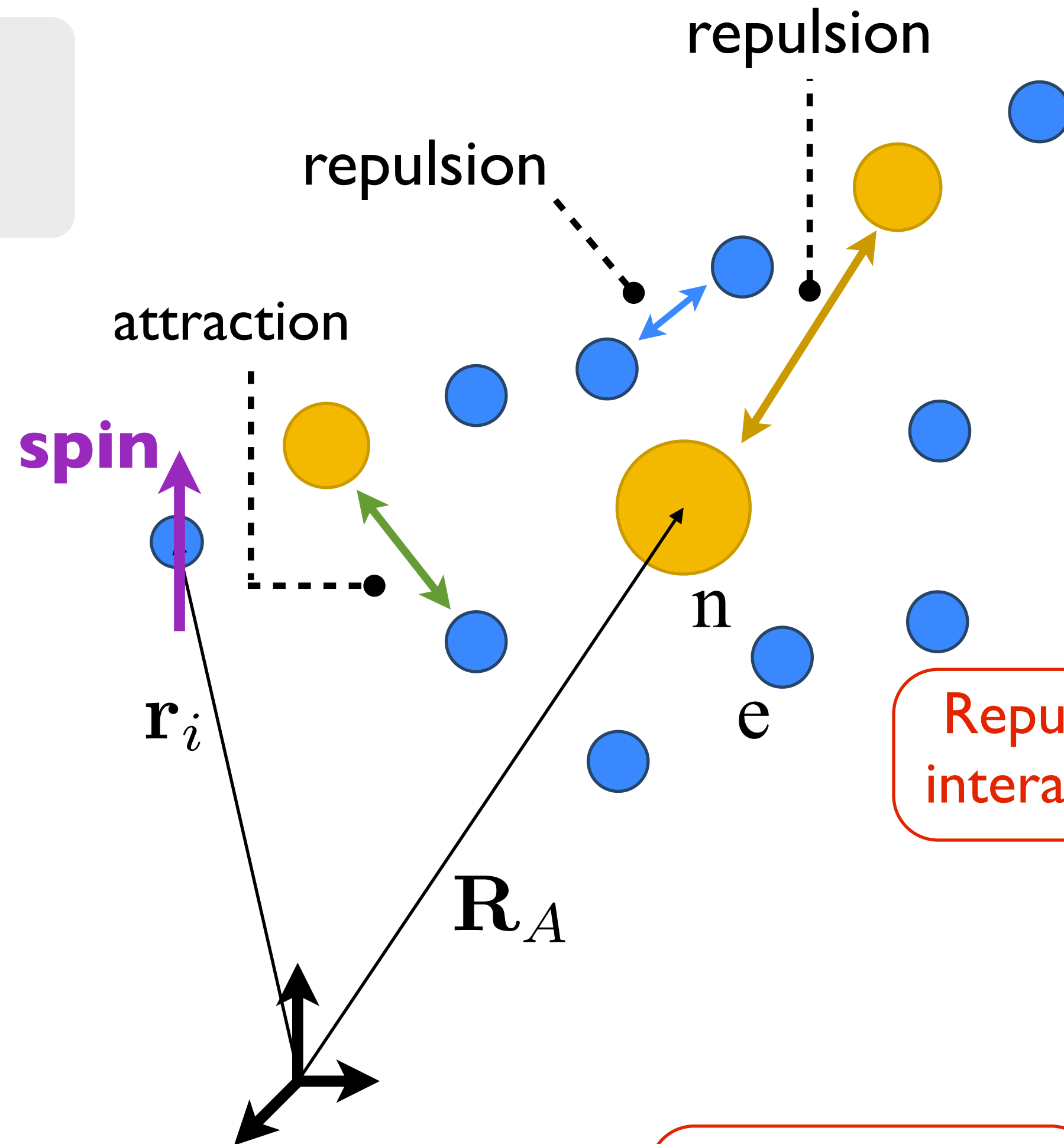
$$\underline{\mathbf{x}} = \{\mathbf{x}_i\}_{i=1}^{N=10}$$

$$\mathbf{x}_i = (\mathbf{r}_i, s_i)$$

within the Born-Oppenheimer Approximation

Hamiltonian

e = electron
n = nucleus



Repulsive
interaction

A constant
(for fixed nuclei)

Atomic units: $e = m = \hbar = 1$

$$\hat{H} = \hat{T}_e + \hat{V}_{ee} + \hat{V}_{ne} + \hat{V}_{nn}$$

$$\hat{T}_e = - \sum_i \frac{\nabla_i^2}{2}$$

Kinetic operator

$$\hat{V}_{ee} = \frac{1}{2} \sum_{i \neq j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$

$$\hat{V}_{en} = \sum_i \sum_A \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|}$$

External potential

$$\hat{V}_{nn} = \frac{1}{2} \sum_{A \neq B} \frac{Z_A Z_B}{|\mathbf{R}_A - \mathbf{R}_B|}$$

State of electrons = solution of Schrödinger equation

$$\hat{H}\Psi_E = E\Psi_E$$

Electronic wavefunction (BOA)

$$\Psi_{\underline{\mathbf{R}}}(\underline{\mathbf{x}})$$

$$\underline{\mathbf{R}} = \{\mathbf{R}_A\}^M$$

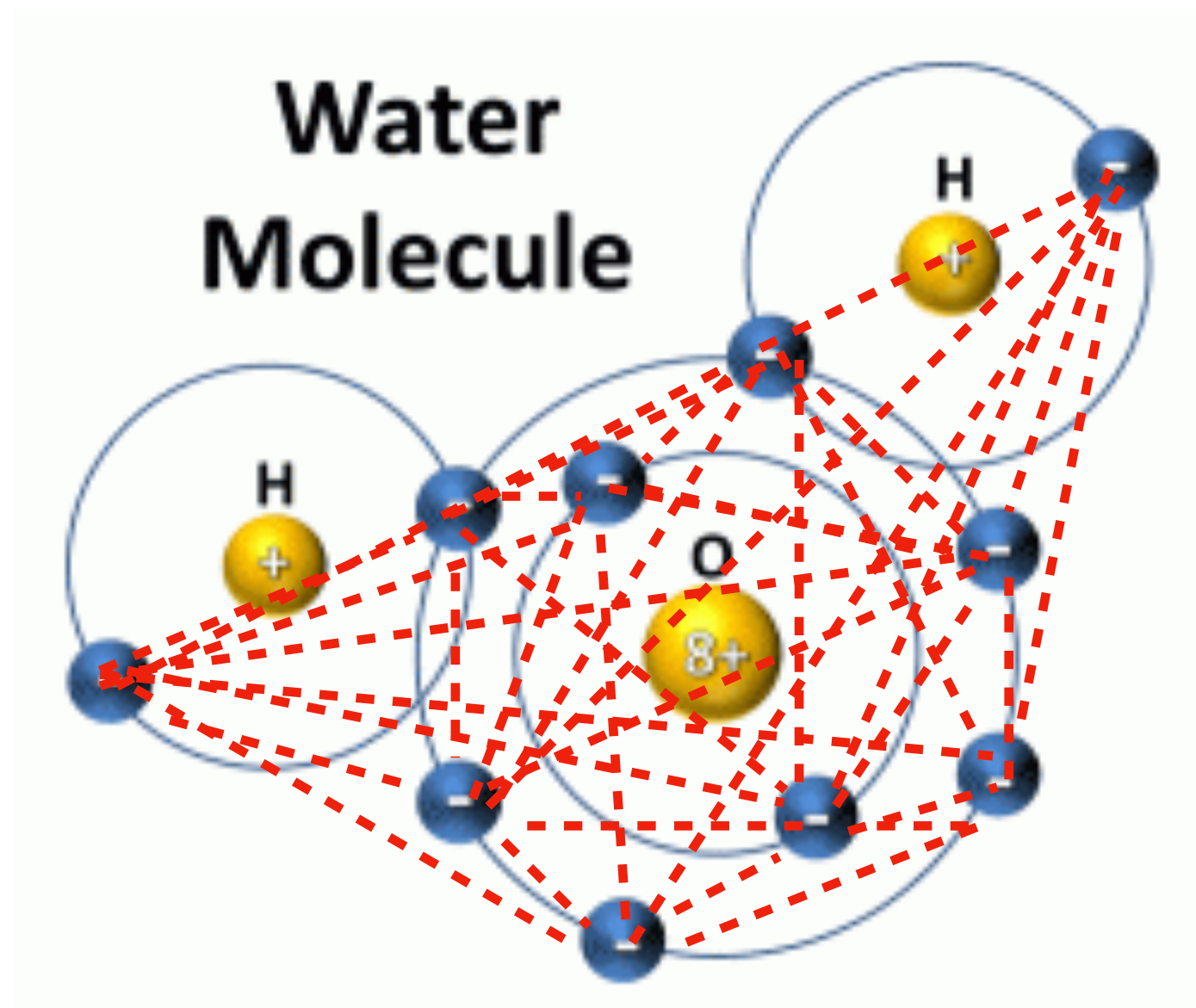
$$\underline{\mathbf{x}} = \{\mathbf{r}_i, s_i\}$$

But we cannot solve it exactly...

not even for a few electrons!

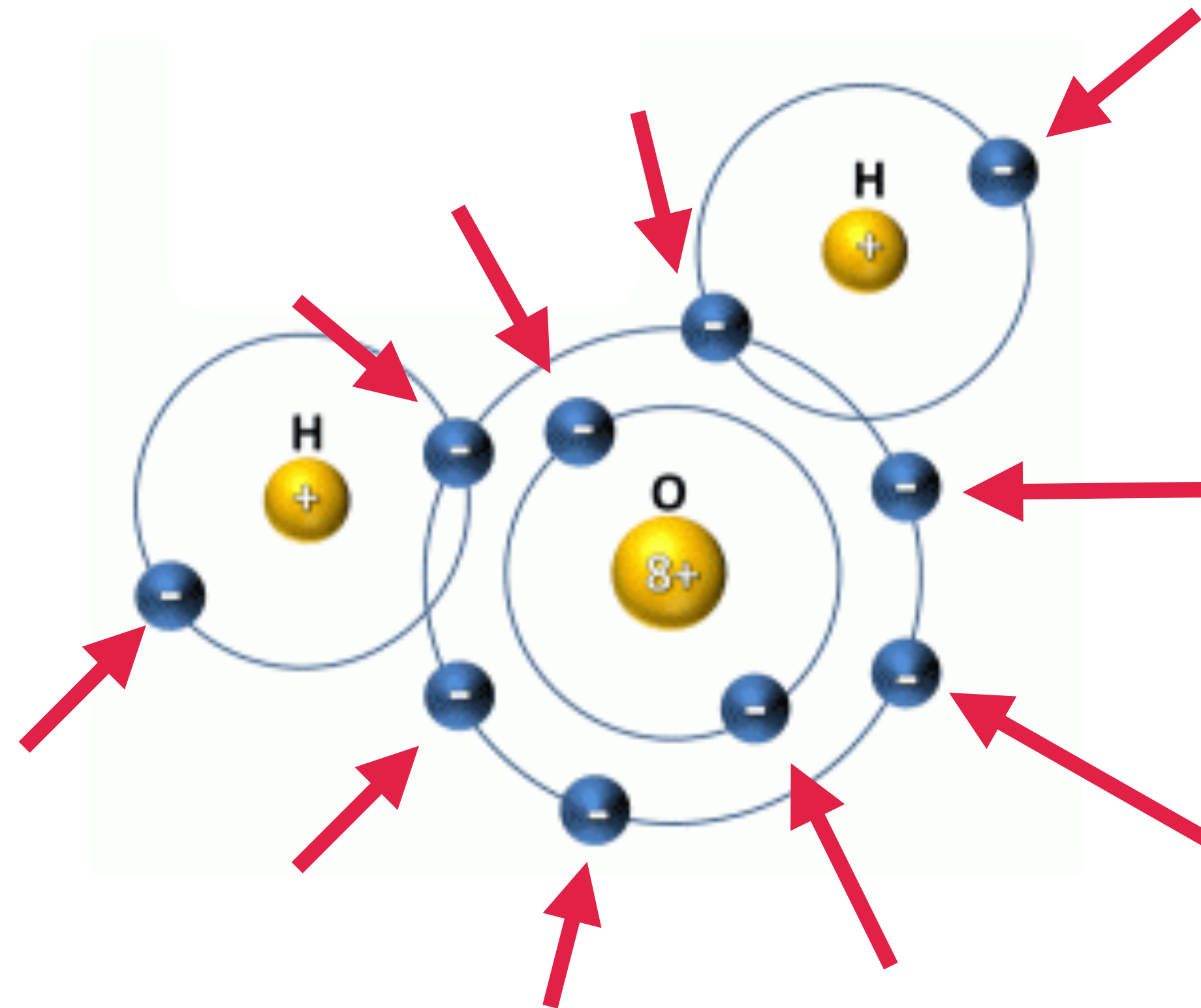
Difficulty 1: electron-electron **interaction**

$$\left[\hat{T} + \hat{V}_{\text{ext}} + \hat{V}_{\text{ee}} \right] |\Psi_{\kappa}\rangle = E_{\kappa} |\Psi_{\kappa}\rangle$$



$$|\Psi_0\rangle =$$

Difficulty 2: **many** electrons



State of electrons = solution of Schrödinger equation

$$\hat{H}\Psi_E = E\Psi_E$$

Electronic wavefunction

Ψ

Need reformulations...

$$\underline{\mathbf{R}} = \{\mathbf{R}_A\}_{A=1}^{M=3}$$

$$\underline{\mathbf{x}} = \{\mathbf{x}_i\}_{i=1}^{N=10}$$

$$\mathbf{x}_i = (\mathbf{r}_i, s_i)$$

Alternative to solving diff. eq.s

Ground state energy can be obtained from the minimization of a functional

$$\begin{aligned} E_0 &= \min_{\Psi} \langle \Psi | \hat{H} | \Psi \rangle \\ &= \min_{\Psi} E[\Psi] \end{aligned}$$

Anti-symmetric wavefunction
for N electrons satisfying:

- proper boundary conditions
- normalization

$$\langle \Psi | \Psi \rangle = 1$$

Our first functional!

Alternative to solving diff. eq.s

Ground state energy can be obtained from the **minimum** of a functional

$$E_0 = \min_{\Psi} \langle \Psi | H | \Psi \rangle$$

Let's reformulate it in terms
of simpler quantities

Our first functional!

$$\langle \Psi | \Psi \rangle = 1$$

DFT

Particle density

$$\rho(\mathbf{r}) = N \sum_{\sigma} \int d\mathbf{x}_2 \dots d\mathbf{x}_N |\Psi(\mathbf{r}\sigma, \mathbf{x}_2, \dots, \mathbf{x}_N)|^2$$

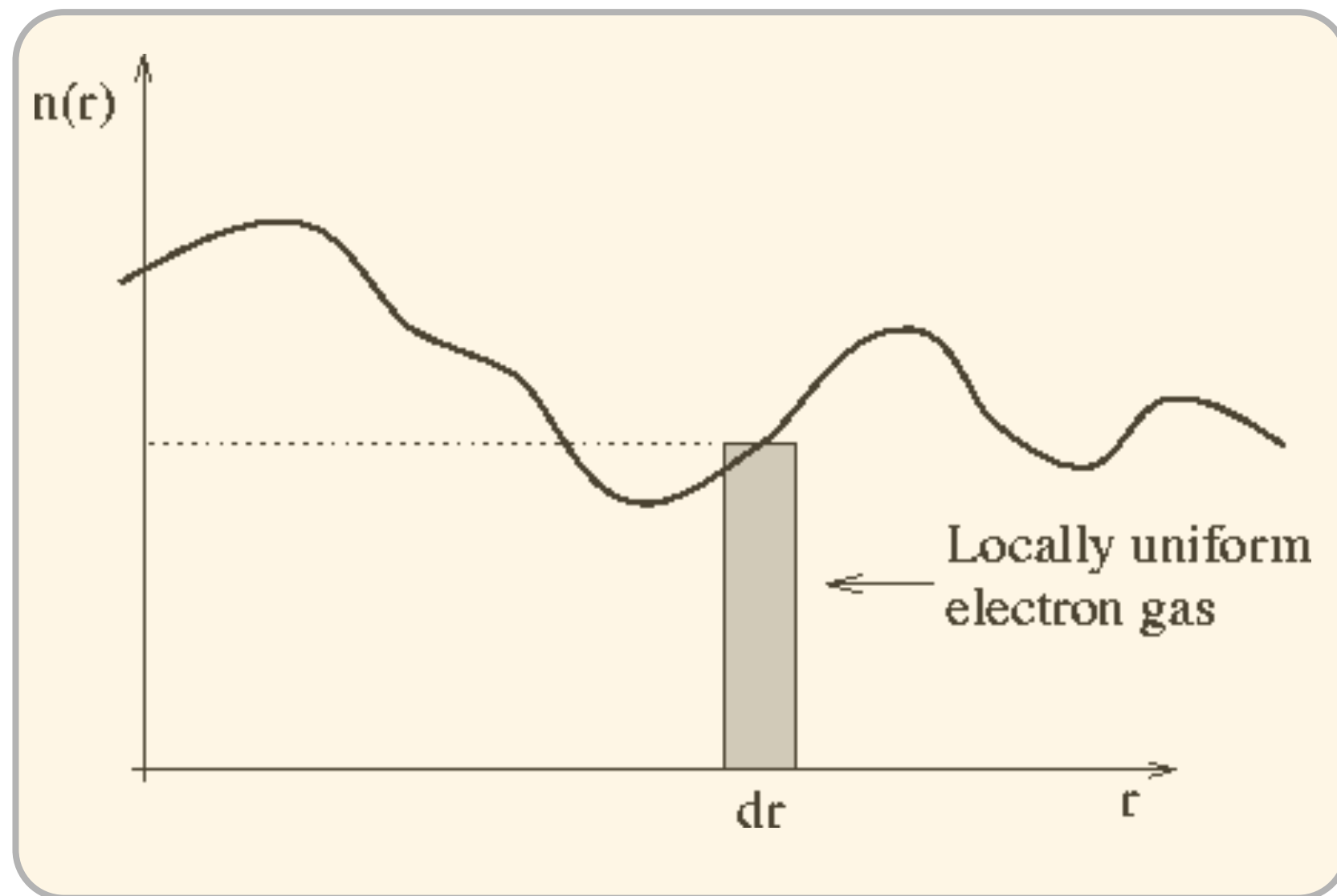
Given $\rho(\mathbf{r})$ we can obtain the energy due to the external potential

$$\langle \Psi | \hat{V}_{\text{ext}} | \Psi \rangle = \int d^3r \, v_{\text{ext}}(\mathbf{r}) \rho(\mathbf{r})$$

$$\hat{V}_{\text{en}} = \sum_i \sum_A \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|}$$

Thomas-Fermi approximation

$$E_{\text{TF}}[n] = \int d^3r \frac{3 (3\pi^2)^{2/3}}{10} \rho^{5/3}(\mathbf{r}) + \frac{1}{2} \int d^3r \int d^3r' \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + \int d^3r (v_{\text{ext}}(\mathbf{r}) - \mu) \rho(\mathbf{r})$$



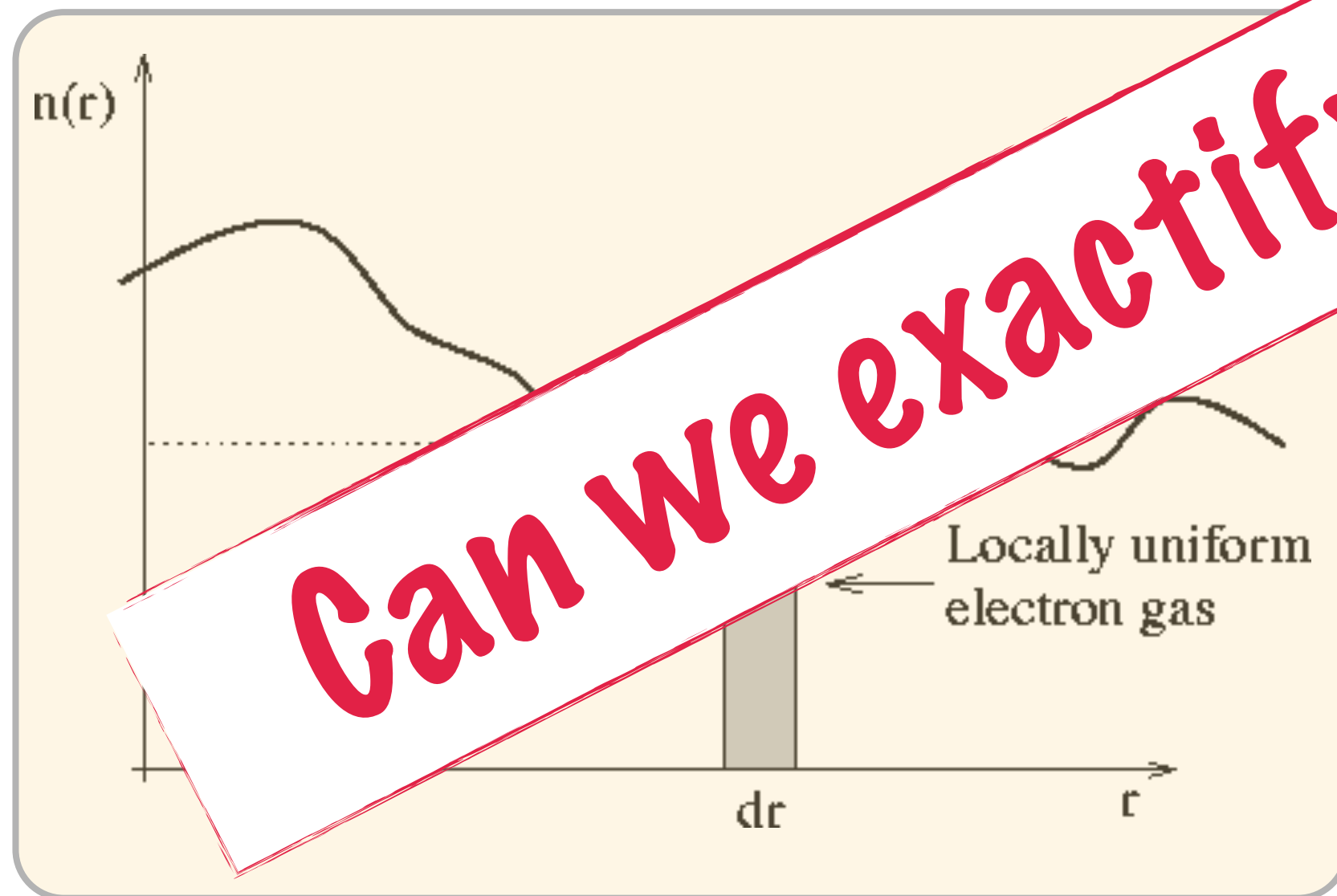
Semiclassical approximation

pairs distributed independently

*wrong density, no quantum shells,
no molecular binding*

Thomas-Fermi approximation

$$E_{\text{TF}}[n] = \int d^3r \frac{3 (3\pi^2)^{2/3}}{10} \rho^{5/3}(\mathbf{r}) + \frac{1}{2} \int d^3r \int d^3r' \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} - \mu \int d^3r \rho(\mathbf{r})$$



Can we exactify it? (in some sense)

Semiclassical approximation

pairs distributed independently

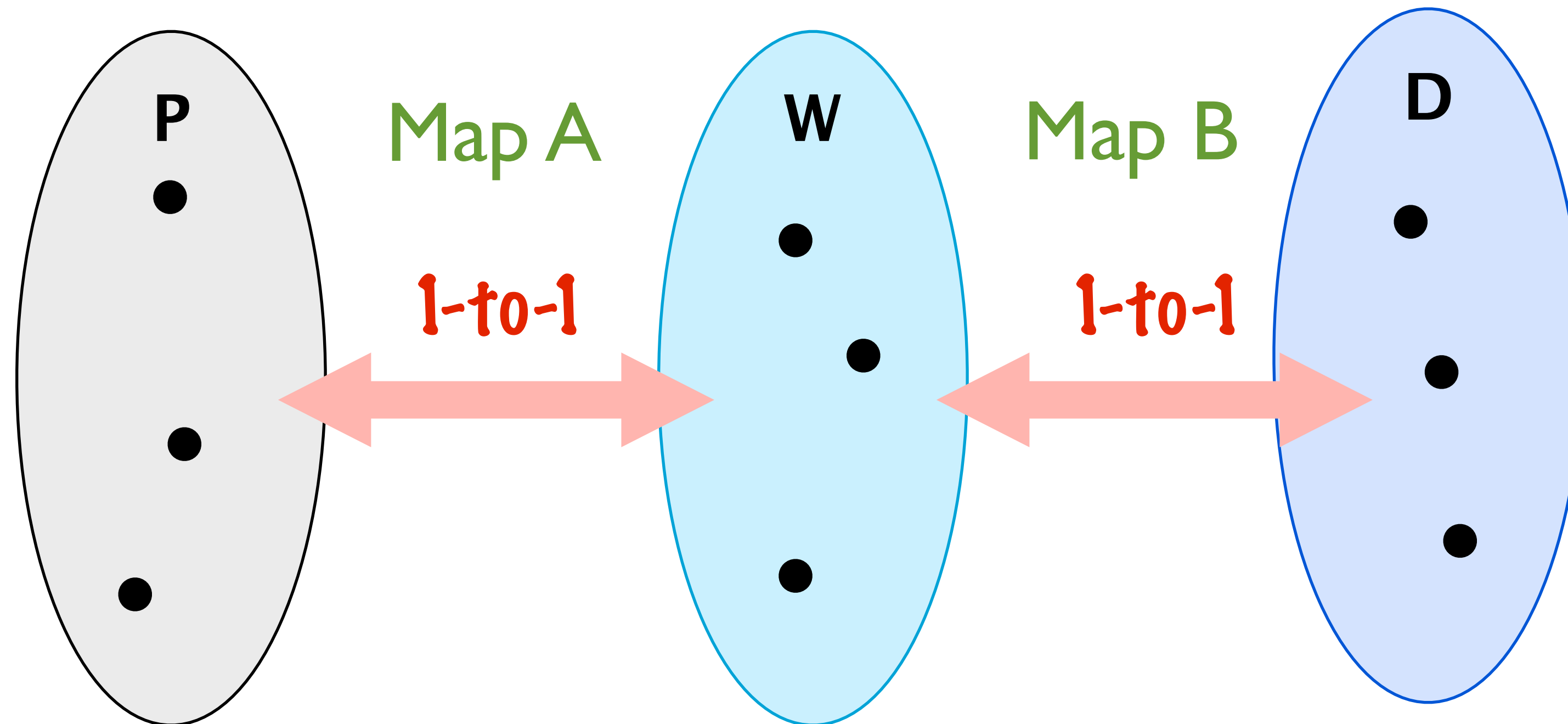
*wrong density, no quantum shells,
no molecular binding*

Hohenberg-Kohn (HK) Theorem

Let us consider an N -electron system with **non-degenerate**(*) ground state (gs)...

P: Set of external (local) potentials
with non-deg. gs

D: Set of v -representable
gs particle density



W: Set of gs wavefunctions

Observables as density functionals

All ground state observables are functional of $\rho(\mathbf{r})$

$$O[\rho] = \langle \Psi[\rho] | \hat{O} | \Psi[\rho] \rangle$$

An important example

$$E[\rho] = \langle \Psi[\rho] | \hat{H} | \Psi[\rho] \rangle = \langle \Psi[\rho] | \hat{T} + \hat{V}_{\text{ee}} | \Psi[\rho] \rangle + \int d^3r v_{\text{ext}}(\mathbf{r}) \rho(\mathbf{r})$$



$$F[\rho] = \langle \Psi[\rho] | \hat{T} + \hat{V}_{\text{ee}} | \Psi[\rho] \rangle$$

“universal” functional

Lyrics:

All you need is ρ , ρ ...
It's easy





$n(\mathbf{r})$

**Hohenberg & Kohn
1964**

Variational principle: reformulated!

From wavefunctions

$$E_{v_{\text{ext}}} := \min_{\Psi} \langle \Psi | \hat{T} + \hat{V}_{\text{ee}} + \hat{V}_{\text{ext}} | \Psi \rangle$$

To particle densities

$$E_{v_{\text{ext}}} := \min_n \left\{ F[n] + \int d^3r \, v_{\text{ext}}(\mathbf{r}) n(\mathbf{r}) \right\}$$

Variational principle: reformulated!

From wavefunctions

$$E_{v_{\text{ext}}} := \min_{\Psi} \langle \Psi | \hat{T} + \hat{V}_{\text{ext}} | \Psi \rangle$$

From particle densities

$$E_{v_{\text{ext}}} := \min_n \left\{ F[n] + \int d^3r \, v_{\text{ext}}(\mathbf{r}) n(\mathbf{r}) \right\}$$

In principle! And in practice?



ALL YOU NEED IS $n(r)$

+ a simple wavefunction

PARLOPHONE

NZP.3236

(7XCE.18425)

NORTHERN
SONGS

Kohn & Sham
1965

Simple wavefunction = (single) Slater determinant

Single-particle state

Spin-orbital

$$\phi(\mathbf{r}, s) = \varphi(\mathbf{r})\chi_\sigma(s)$$

Spin-eigenfunction

$$\chi_\uparrow = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \chi_\downarrow = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

For non-degenerate gs

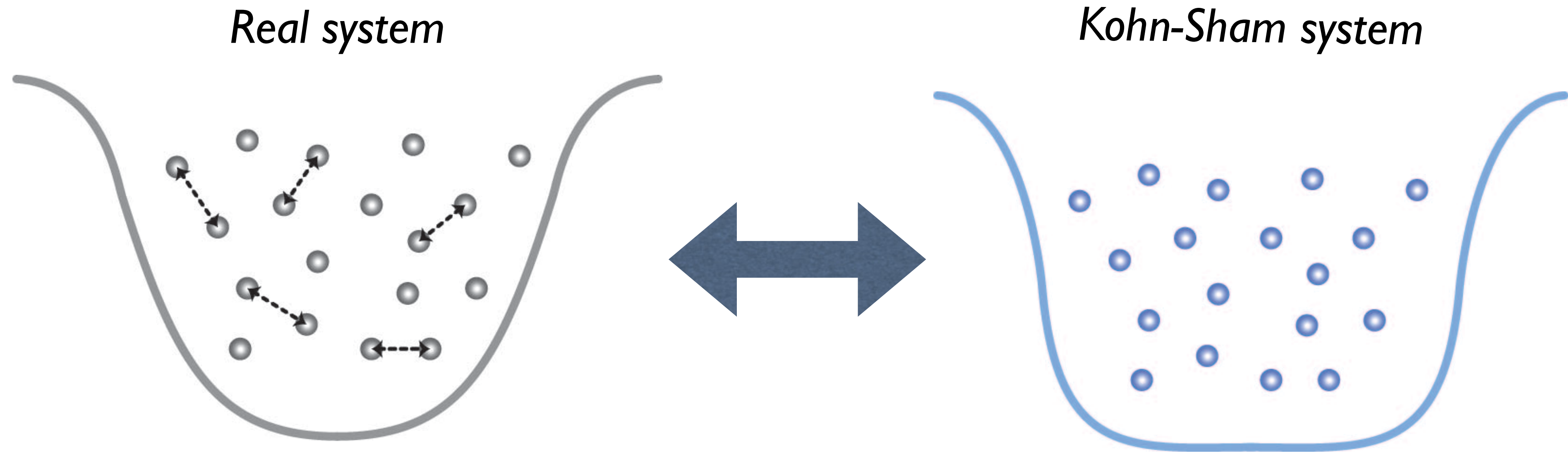
$$N_\uparrow = N_\downarrow$$

Anti-symmetric N -particle state

$$\Phi_s(\mathbf{x}_1, \dots, \mathbf{x}_N) = \frac{1}{\sqrt{N!}} \det \begin{bmatrix} \phi_1(\mathbf{r}_1, s_1) & \phi_2(\mathbf{r}_1, s_1) & \dots & \phi_N(\mathbf{r}_1, s_1) \\ \phi_1(\mathbf{r}_2, s_2) & \phi_2(\mathbf{r}_2, s_2) & \dots & \phi_N(\mathbf{r}_2, s_2) \\ \vdots & \vdots & \vdots & \vdots \\ \phi_1(\mathbf{r}_N, s_N) & \phi_2(\mathbf{r}_N, s_N) & \dots & \phi_N(\mathbf{r}_N, s_N) \end{bmatrix}$$

orthonormal orbitals

Kohn-Sham systems



Interacting electrons
in a given
external potential

$$\hat{H}\Psi_{\text{gs}} = E_{\text{gs}}\Psi_{\text{gs}}$$

**Let us restrict to
systems with non-degenerate gs**

Non-interacting electrons
in a **different** potential
but with **same** ground-state density
as in the real system

$$\left\{ -\frac{1}{2}\nabla^2 + v_s \right\} \varphi_{j\sigma} = \epsilon_k \varphi_{j\sigma}$$

**The KS state is a single Slater
determinant**

Kohn-Sham (KS) formulation of DFT (1/2)

Exact (ground-state) energy density functional

$$E[n] = T_s[n] + E_H[n] + E_{xc}[n] + \int d^3r v_{\text{ext}}(\mathbf{r})n(\mathbf{r})$$

By definition

Kinetic energy of the KS system

$$T_s[n] = - \sum_{\sigma} \sum_{j=1}^{N_{\sigma}} \int d^3r \frac{\varphi_{j\sigma}^*(\mathbf{r}) \nabla^2 \varphi_{j\sigma}(\mathbf{r})}{2}$$

$$\varphi_{j\sigma} = \varphi_{j\sigma}[n]$$

$$E_H[n] = \frac{1}{2} \int d^3r \int d^3r' \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}$$

Hartree energy

$$E_{xc}[n] := F[n] - \left(T_s[n] + E_H[n] \right)$$

*Exchange-correlation energy
(must be approximated!)*

Kohn-Sham (KS) formulation of DFT (2/2)

$$\left\{ -\frac{1}{2}\nabla^2 + v_s(\mathbf{r}) \right\} \varphi_{j\sigma}(\mathbf{r}) = \epsilon_k \varphi_{j\sigma}(\mathbf{r})$$

$$\begin{aligned} \varphi_{j\uparrow} &= \varphi_{j\downarrow} \\ N_{\uparrow} &= N_{\downarrow} \end{aligned}$$

$$v_s(\mathbf{r}) = v_{\text{ext}}(\mathbf{r}) + \int d^3r' \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + v_{\text{xc}}(\mathbf{r})$$

$$v_{\text{xc}} = v_{\text{xc}}[n]$$

$$v_{\text{xc}}(\mathbf{r}) = \frac{\delta E_{\text{xc}}}{\delta n(\mathbf{r})}$$

Exchange-correlation potential

$$n(\mathbf{r}) = \sum_{\sigma} \sum_{j=1}^{N_{\sigma}} |\varphi_{j\sigma}(\mathbf{r})|^2$$

Constraint search formulation

$$E = \min_{\Psi} \langle \Psi | \hat{T} + \hat{V}_{ee} + \hat{V}_{\text{ext}} | \Psi \rangle$$

DFT without HK

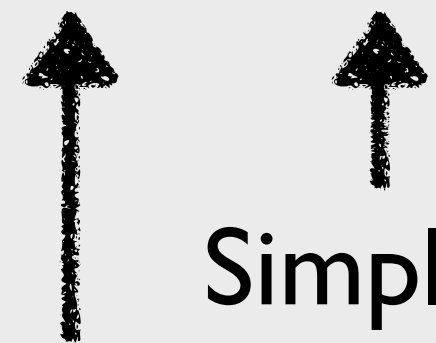
$$(*) = \min_n \left\{ \min_{\Psi \rightarrow n} \langle \Psi | \hat{T} + \hat{V}_{ee} + \hat{V}_{\text{ext}} | \Psi \rangle \right\}$$



$$= \min_n \left\{ \min_{\Psi \rightarrow n} \langle \Psi | \hat{T} + \hat{V}_{ee} | \Psi \rangle + \int d^3r \, n(\mathbf{r}) v_{\text{ext}}(\mathbf{r}) \right\} = \min_n \left\{ F[n] + \int d^3r \, n(\mathbf{r}) v_{\text{ext}}(\mathbf{r}) \right\}$$

$$= \min_n \left\{ \min_{\Phi \rightarrow n} \langle \Phi | \hat{T} + \hat{V}_{ee} | \Phi \rangle + E_c[n] + \int d^3r \, n(\mathbf{r}) v_{\text{ext}}(\mathbf{r}) \right\}$$

KS DFT



Simple wf at the price of introducing the (x)c energy func.

Generates multiplicative local effective potentials

(*) Invoke **N**-representable densities

Levy, Proc. Natl. Acad. Sci. USA 76 6062 (1979)

Lieb, Int. J. Quantum Chem 24, 243 (1983)

Generalized KS system

$$E = \min_{\Psi} \langle \Psi | \hat{T} + \hat{V}_{ee} + \hat{V}_{\text{ext}} | \Psi \rangle$$

DFT without HK

$$= \min_n \left\{ \min_{\Psi \rightarrow n} \langle \Psi | \hat{T} + \hat{V}_{ee} + \hat{V}_{\text{ext}} | \Psi \rangle \right\}$$



$$= \min_n \left\{ \min_{\Psi \rightarrow n} \langle \Psi | \hat{T} + \hat{V}_{ee} | \Psi \rangle + \int d^3r \, n(\mathbf{r}) v_{\text{ext}}(\mathbf{r}) \right\} = \min_n \left\{ F[n] + \int d^3r \, n(\mathbf{r}) v_{\text{ext}}(\mathbf{r}) \right\}$$

$$= \min_n \left\{ \min_{\Phi \rightarrow n} \langle \Phi | \hat{T} + \hat{V}_{ee} | \Phi \rangle + E_c[n] + \int d^3r \, n(\mathbf{r}) v_{\text{ext}}(\mathbf{r}) \right\}$$

KS DFT

$$= \min_{\Phi} \left\{ \langle \Phi | \hat{T} + \hat{V}_{ee} + \hat{V}_{\text{ext}} | \Phi \rangle + E_c[n[\Phi]] \right\}$$

Generalized KS

Generate “non local” effective pot.s

KS Phys. Rev. 140, A1133 (1965)

A. Seidl et al., PRB 53 (1996)

Understanding band gaps of solids in generalized Kohn–Sham theory

John P. Perdew^{a,b,1}, Weitao Yang^c, Kieron Burke^{d,e}, Zenghui Yang^{a,2}, Eberhard K. U. Gross^f, Matthias Scheffler^{g,h,i}, Gustavo E. Scuseria^{j,k}, Thomas M. Henderson^{j,k}, Igor Ying Zhang^g, Adrienn Ruzsinszky^a, Haowei Peng^a, Jianwei Sun^l, Egor Trushin^m, and Andreas Görling^m

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Contributed by John P. Perdew, February 2, 2017 (sent for review December 30, 2016; reviewed by Roi Baer and Stephan Kuemmel)

Most of the xc “discontinuity” is then adsorbed into the G-KS gap.

See also A. Seidl *et al.*, PRB 53 (1996)

A bird's-eye view of density functional approximations

Hierarchy of DFT approx.s



MBP-diagrams:

Hybrids:

Meta-GGAs:

GGAs:

LDAs:

Unoccupied orbitals / eigenvalues

$$E_{xc} \approx \alpha E_x^{\text{Fock}} + (1 - \alpha) E_x^{\text{DFA}} + E_c^{\text{DFA}}$$

$$E_{xc}[n] \approx \int d^3r \epsilon_{xc}(n(\mathbf{r}), \nabla n(\mathbf{r}), \tau(\mathbf{r}))$$

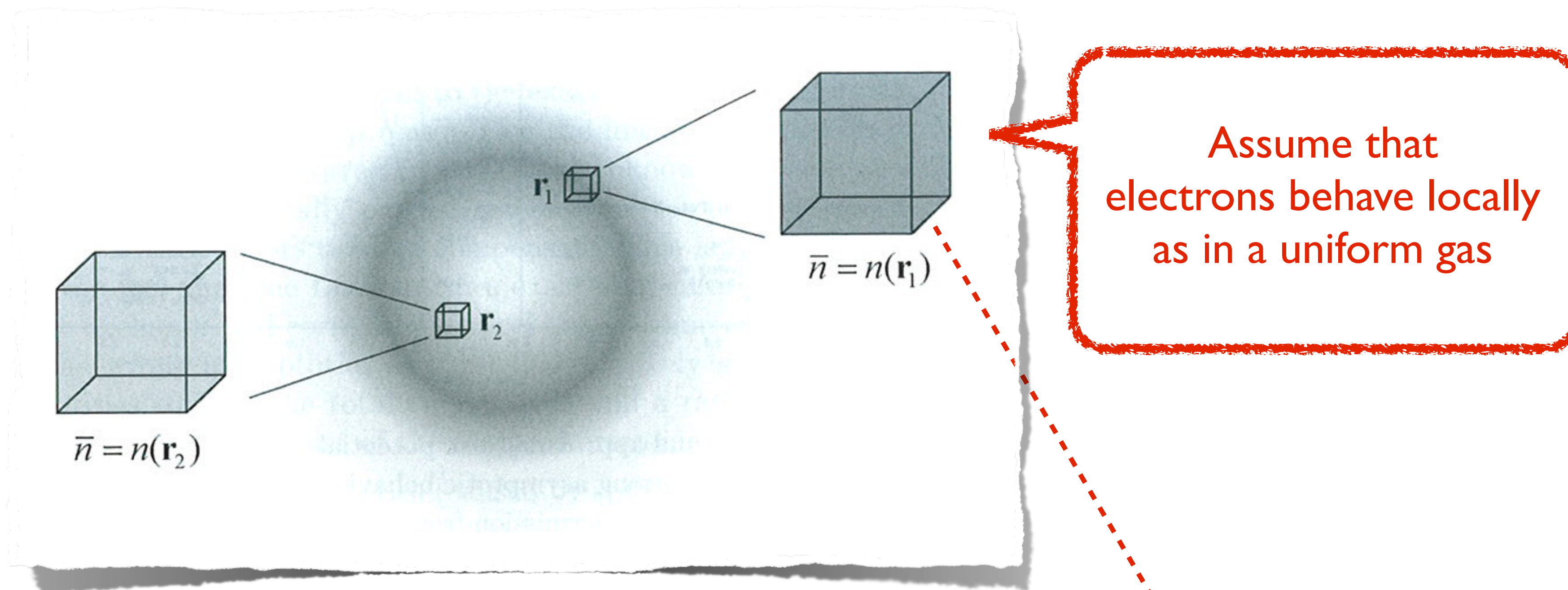
$$E_{xc}[n] \approx \int d^3r \epsilon_{xc}(n(\mathbf{r}), \nabla n(\mathbf{r}))$$

$$E_{xc}[n] \approx \int d^3r \epsilon_{xc}(n(\mathbf{r}))$$

$$\tau(\mathbf{r}) = \frac{1}{2} \sum_{\sigma} \sum_j |\nabla \varphi_{j\sigma}(\mathbf{r})|^2$$

$$E_x^{\text{Fock}} = - \sum_{\sigma} \sum_j \iint d^2r d^3r' \frac{|\sum_i \varphi_{j\sigma}^*(\mathbf{r}) \varphi_{j\sigma}(\mathbf{r}')|^2}{2|\mathbf{r} - \mathbf{r}'|}$$

Local Density Approximation (1/2)



$$E_{\text{xc}}^{\text{LDA}}[n] = \int d^3r \, n(\mathbf{r}) \epsilon_{\text{xc}}^{\text{unif}}(n(\mathbf{r}))$$

Local Density Approximation (2/2)

- **Uniform Electron Gas (UEG):** constant electron density that fills the entire universe; the negative charge of the electrons is neutralized by a rigid uniform positive background; periodic boundary condition (for a cube of volume growing to infinity);
- **Many Body Perturbation Theory** and **Quantum Monte Carlo** provide us with accurate, if not exact, detailed information.

Hierarchy of DFT approx.s



MBP-diagrams:

Hybrids:

Meta-GGAs:

GGAs:

LDAs:

Unoccupied orbitals / eigenvalues

$$E_{xc} \approx \alpha E_x^{\text{Fock}} + (1 - \alpha) E_x^{\text{DFA}} + E_c^{\text{DFA}}$$

$$E_{xc}[n] \approx \int d^3r \epsilon_{xc}(n(\mathbf{r}), \nabla n(\mathbf{r}), \tau(\mathbf{r}))$$

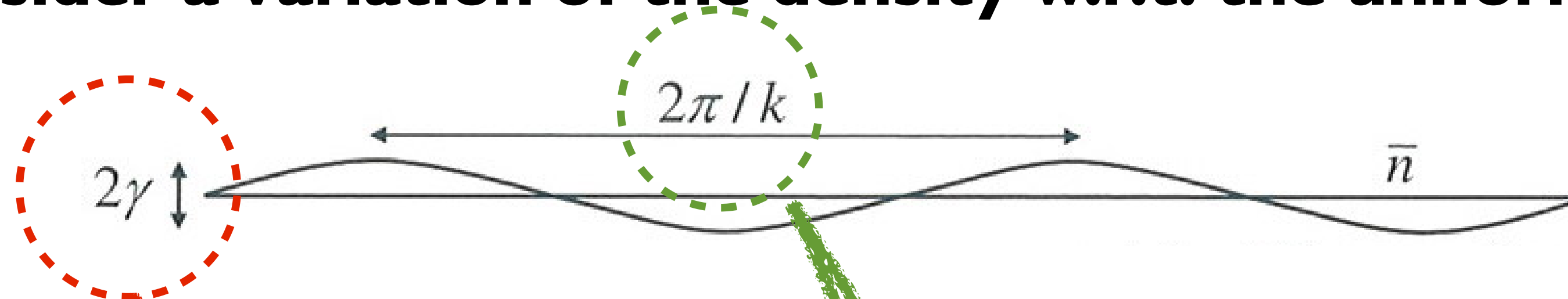
$$E_{xc}[n] \approx \int d^3r \epsilon_{xc}(n(\mathbf{r}), \nabla n(\mathbf{r})) \quad \leftarrow$$

$$E_{xc}[n] \approx \int d^3r \epsilon_{xc}(n(\mathbf{r}))$$

GGA stands for **Generalized** Gradient Approximation

Gradient Expansion Approximation (1/3)

Consider a variation of the density w.r.t. the uniform state



Variation is small if $\frac{|\delta n|}{n_0} \ll 1$

Fermi wave number

$$k_F(\mathbf{r}) = [3\pi^2 n(\mathbf{r})]^{1/3}$$

TF screening wave number

$$k_s(\mathbf{r}) = \sqrt{4k_F(\mathbf{r})}$$

Gradient Expansion Approximation (2/3)

For slow density variations, we get

$$E_x[n] = A_x \int d^3r \, n^{4/3} [1 + \mu s^2 + \dots]$$

$$s(\mathbf{r}) = \frac{|\nabla n(\mathbf{r})|}{2k_F(\mathbf{r})n(\mathbf{r})}$$

It applies for $s \ll 1$

*It can be determined
from linear response*

$$E_c[n] = \int d^3r \, [e_c^{\text{unif}}(n) + \beta(n)t^2 + \dots]$$

$$t(\mathbf{r}) = \frac{|\nabla n(\mathbf{r})|}{2k_s(\mathbf{r})n(\mathbf{r})}$$

It applies for $t \ll 1$

*It can be determined
from linear resp. +
high-density limit*

Gradient Expansion Approximation (3/3)

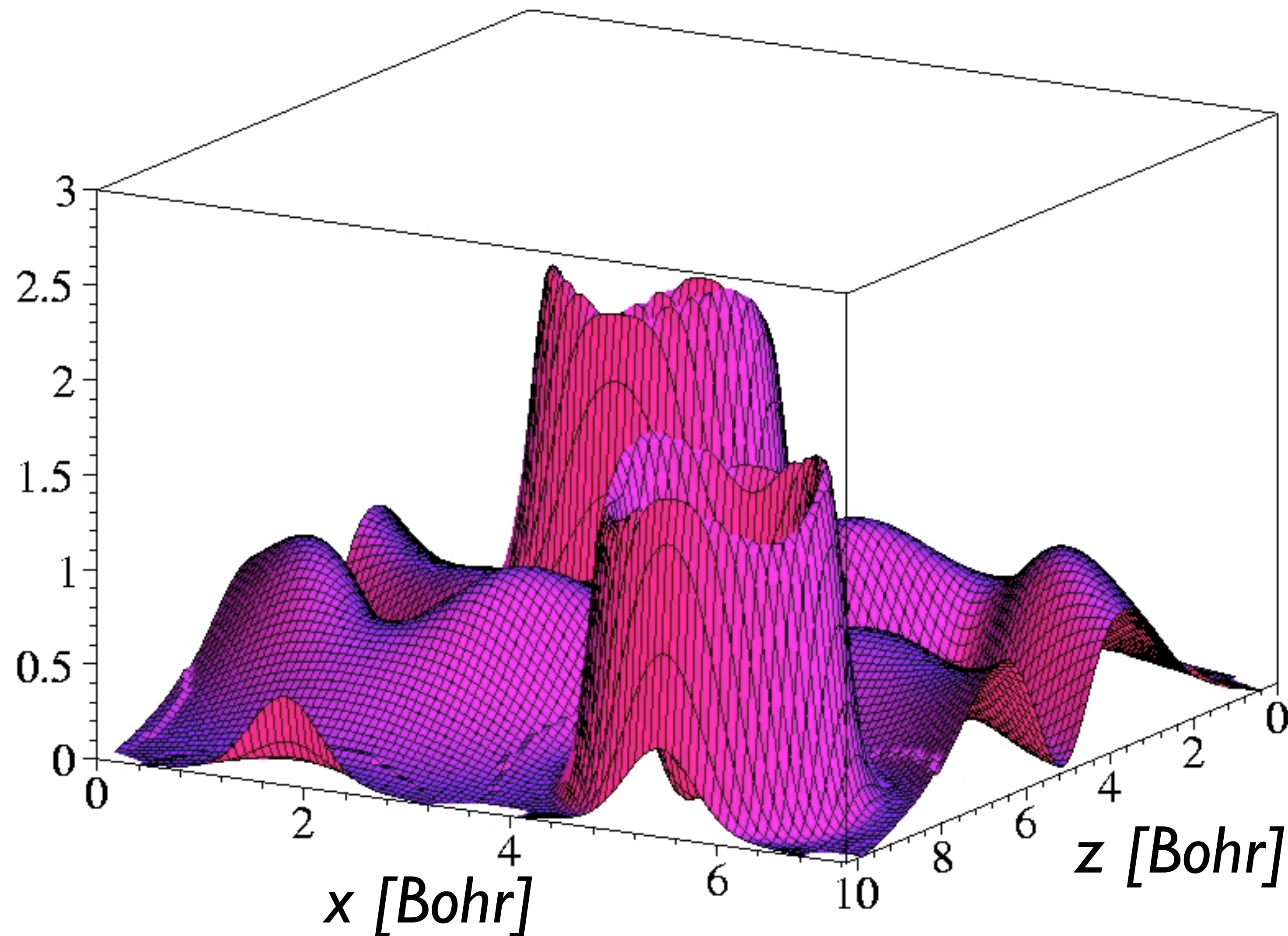
GEAs do **not** improve over LDA...

Both x and c are INADEQUATE

$E_c > 0$ — wrong sign!

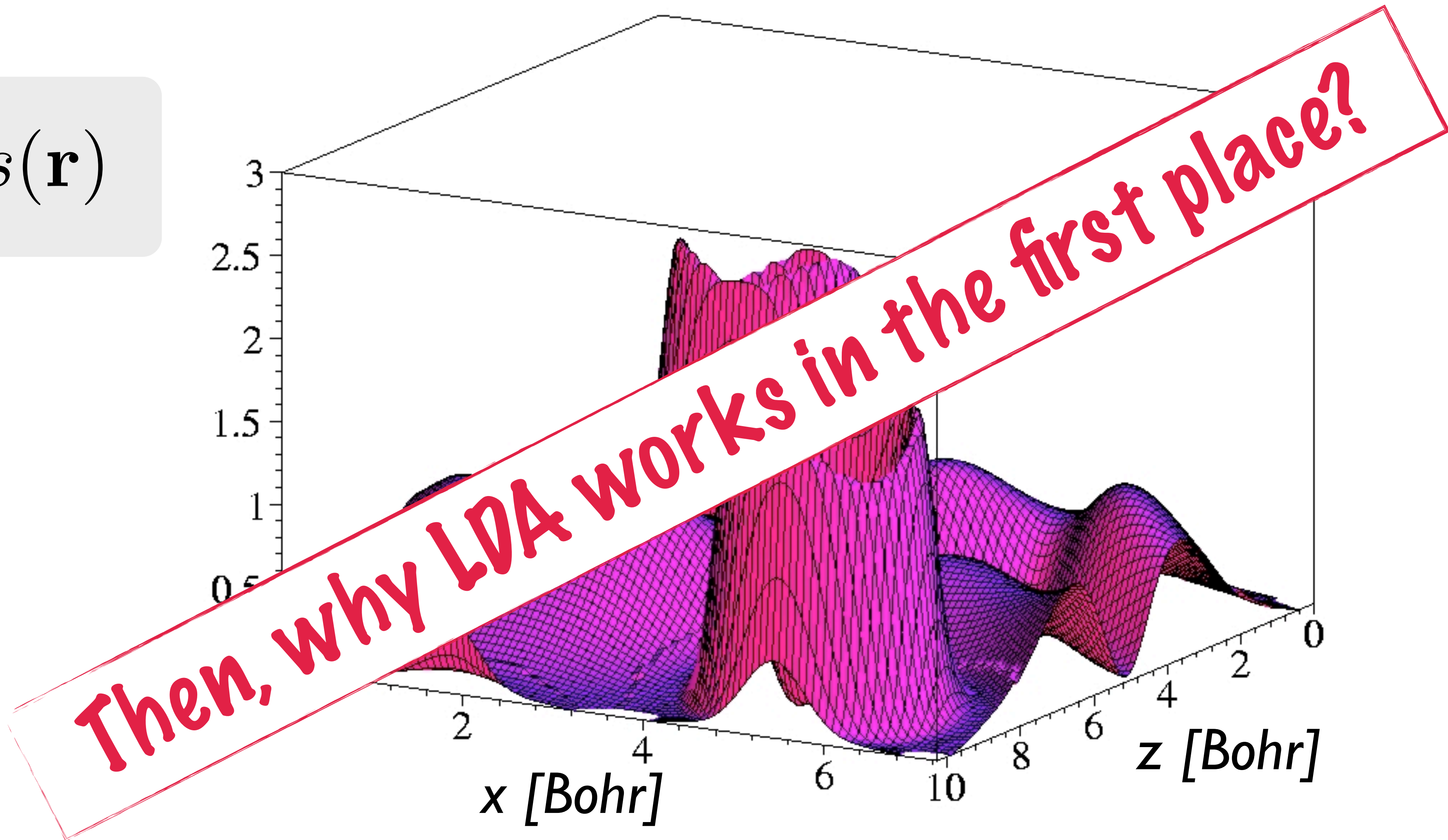
Bulk Silicon (diamond structure)

$$s(\mathbf{r})$$



Bulk Silicon (diamond structure)

$$s(\mathbf{r})$$

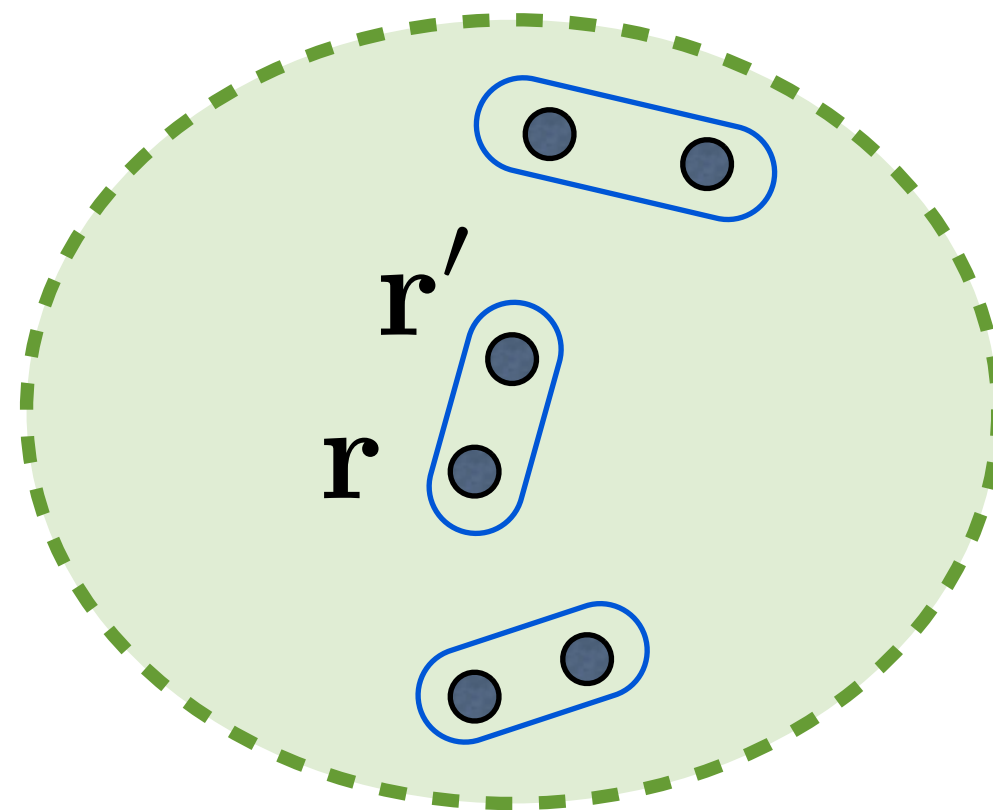


DFT was stuck in a hole!

So, let's give a look at the hole function...

Interaction energy and pairs

Electron-electron interaction



Pair distribution

$$\langle \Psi | \hat{V}_{ee} | \Psi \rangle = \frac{1}{2} \int d^3r \int d^3r' \frac{n_2(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}$$

wavefunction

$$n_2(\mathbf{r}, \mathbf{r}') = N(N-1) \sum_{\sigma, \sigma'} \int dx_3 \dots dx_N |\Psi(\mathbf{r}\sigma, \mathbf{r}'\sigma', \mathbf{x}_3, \dots, \mathbf{x}_N)|^2$$

Hole function

Pair distribution in terms of the particle density and hole function

$$n_2(\mathbf{r}, \mathbf{r}') \equiv n(\mathbf{r}) [n(\mathbf{r}') + h(\mathbf{r}, \mathbf{r}')]]$$

The hole function accounts for -1 electron

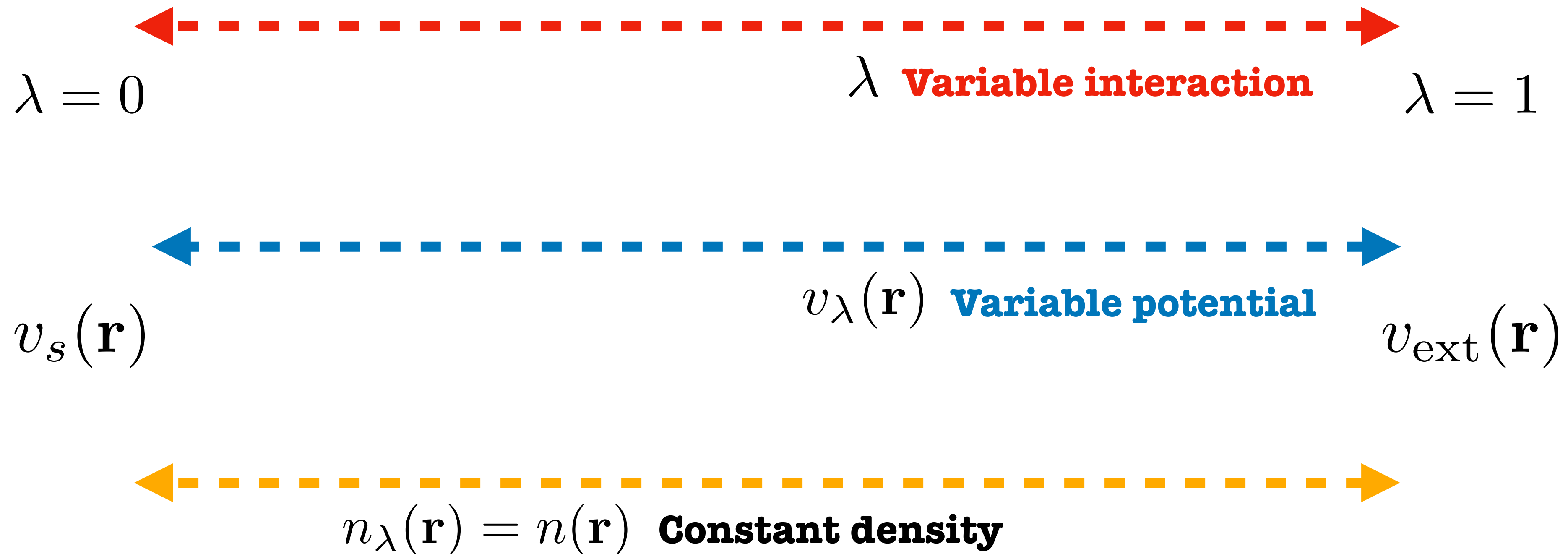
$$\int d^3r' h(\mathbf{r}, \mathbf{r}') = -1$$

Electrons avoid each others

$$\langle \Psi | \hat{V}_{ee} | \Psi \rangle = \frac{1}{2} \int d^3r \int d^3r' \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \quad \text{Hartree (Coulomb) energy}$$
$$+ \frac{1}{2} \int d^3r \int d^3r' \frac{n(\mathbf{r})h(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \quad \text{Fermi-Coulomb-hole energy}$$

Adiabatic Connection

$$\hat{H}_\lambda = \hat{T} + \lambda \hat{V}_{\text{ee}} + \hat{V}_\lambda$$



* Koch and Holthausen "A Chemist's Guide to Density Functional Theory", Second Edition, Wiley-VCH Verlag GmbH

* Becke J. Phys. Chem **92**, 1053 (1988)

x- and c-hole functions in KS-DFT

$$E_x[n] = \frac{1}{2} \int d^3r \int d^3r' \frac{n(\mathbf{r})n_x(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} < 0$$

$$E_c[n] = \frac{1}{2} \int d^3r \int d^3r' \frac{n(\mathbf{r})n_c(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \leq 0$$

The **exchange hole** is defined by:

$$n_x(\mathbf{r}, \mathbf{r}') = - \sum_{\sigma} \frac{|\sum_{i=1}^{N_{\sigma}} \varphi_{i\sigma}^*(\mathbf{r}') \varphi_{i\sigma}(\mathbf{r})|^2}{n(\mathbf{r})} < 0$$

The **correlation hole** is (see, for example, *)

$$n_c(\mathbf{r}, \mathbf{r}') = \int_0^1 h^{\lambda}(\mathbf{r}, \mathbf{r}') d\lambda - n_x(\mathbf{r}, \mathbf{r}')$$

(interaction strength)

it **satisfies**:

$$\int d^3r' n_x(\mathbf{r}, \mathbf{r}') = -1$$

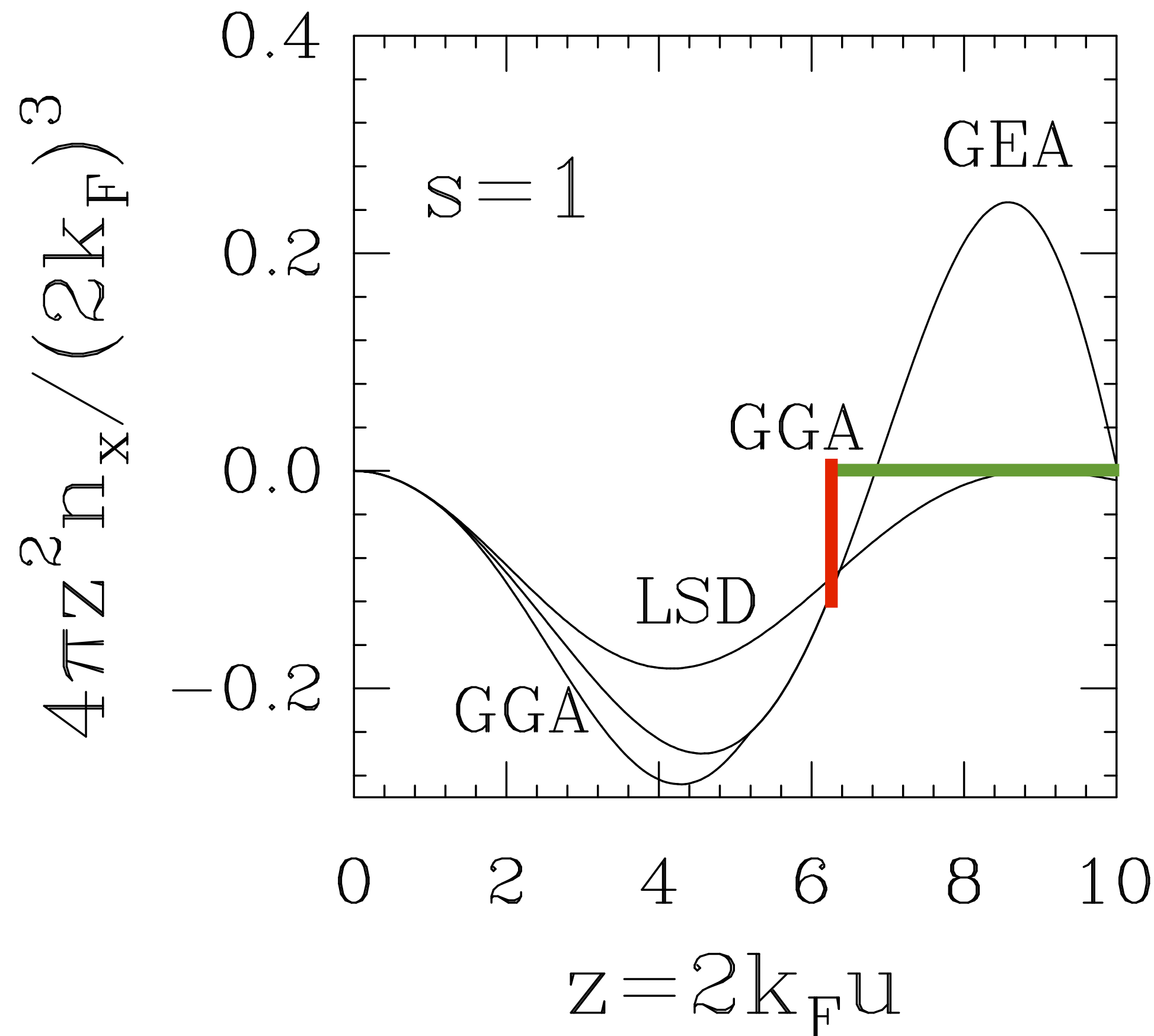
it **satisfies**:

$$\int d^3r' n_c(\mathbf{r}, \mathbf{r}') = 0$$

* Koch and Holthausen "A Chemist's Guide to Density Functional Theory", Second Edition, Wiley-VCH Verlag GmbH

* Becke J. Phys. Chem 88, 1053 (1988)

From GEA to GGA: exchange

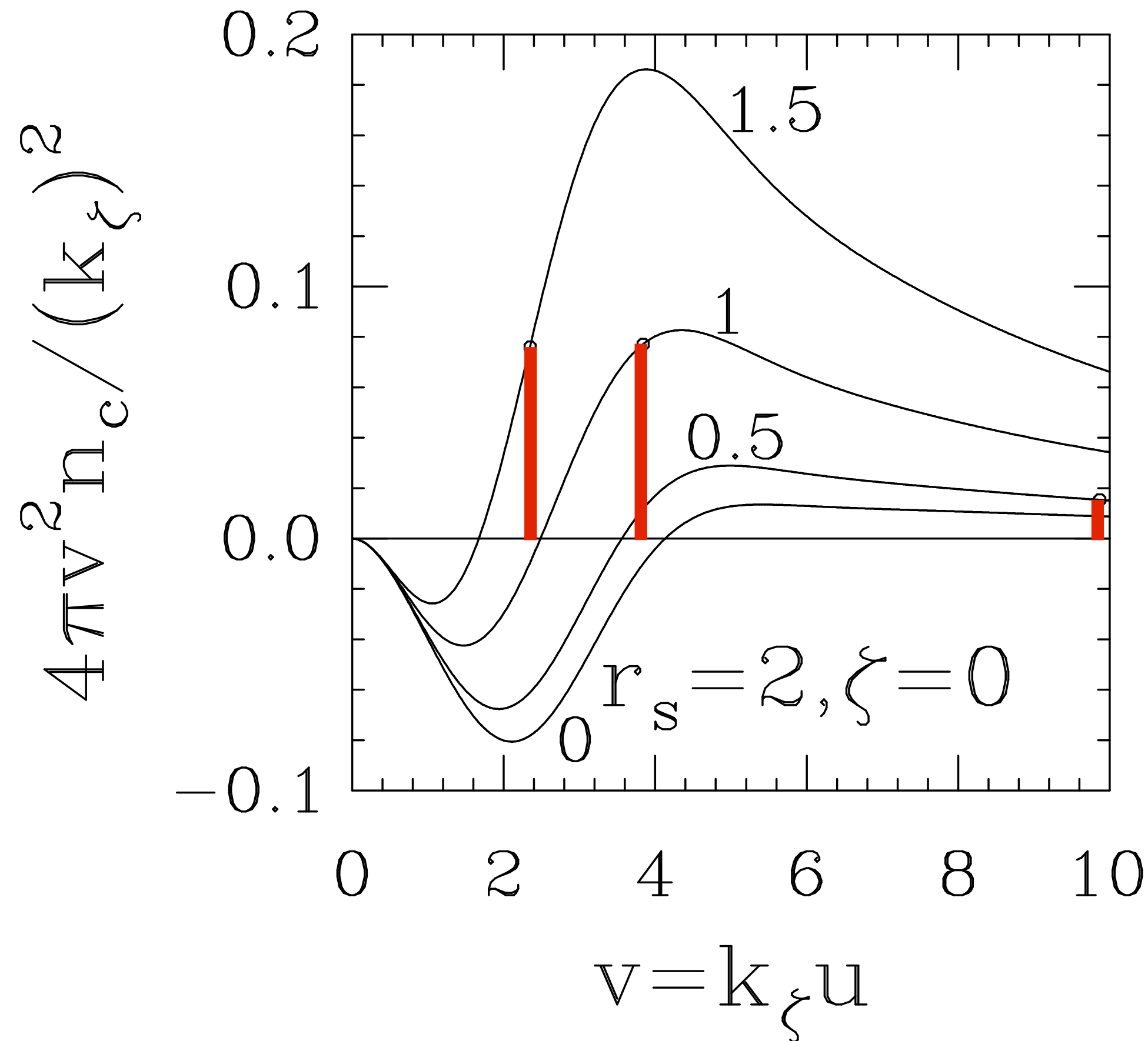


GGA:
makes the GEA x-hole non-positive
and
enforces x-hole normalization

$$\int d^3 r' n_x(\mathbf{r}, \mathbf{r}') = -1$$

Figure 1. Spherically-averaged exchange hole density n_x for $s = 1$.

From GEA to GGA: correlation



GGA:
enforces c-hole normalization
(c-hole may be positive, too)

$$\int d^3r' n_c(\mathbf{r}, \mathbf{r}') = 0$$

Figure 4. Spherically-averaged correlation hole density $\tilde{n}_c$ for $r_s = 2$ and $\zeta = 0$. GEA holes are shown for four values of the reduced density gradient, $t = |\nabla n|/(2k_\zeta n)$. The vertical lines indicate where the numerical GGA cuts off the GEA hole to make $\int_0^{v_c} dv 4\pi v^2 \tilde{n}_c(v) = 0$.

Importance of fulfilling exact conditions (1/2)

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PHYSICAL REVIEW LETTERS

28 OCTOBER 1996

Generalized Gradient Approximation Made Simple

John P. Perdew, Kieron Burke,* Matthias Ernzerhof

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(Received 21 May 1996)

Generalized gradient approximations (GGA's) for the exchange-correlation energy improve upon the local spin density (LSD) description of atoms, molecules, and solids. We present a simple derivation of a simple GGA, in which all parameters (other than those in LSD) are fundamental constants. Only general features of the detailed construction underlying the Perdew-Wang 1991 (PW91) GGA are invoked. Improvements over PW91 include an accurate description of the linear response of the uniform electron gas, correct behavior under uniform scaling, and a smoother potential.

Hierarchy of DFT approx.s



MBP-diagrams:

Hybrids:

Meta-GGAs:

GGAs:

LDAs:

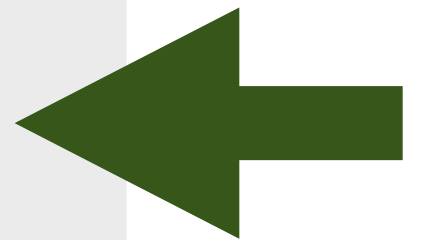
Unoccupied orbitals / eigenvalues

$$E_{xc} \approx \alpha E_x^{\text{Fock}} + (1 - \alpha) E_x^{\text{DFA}} + E_c^{\text{DFA}}$$

$$E_{xc}[n] \approx \int d^3r \epsilon_{xc}(n(\mathbf{r}), \nabla n(\mathbf{r}), \tau(\mathbf{r}))$$

$$E_{xc}[n] \approx \int d^3r \epsilon_{xc}(n(\mathbf{r}), \nabla n(\mathbf{r}))$$

$$E_{xc}[n] \approx \int d^3r \epsilon_{xc}(n(\mathbf{r}))$$



x-hole also helps to understand “Meta-GGAs”

Exact short-range behavior of the x-hole

$$n_x^\sigma(\mathbf{r}, u) = -n_\sigma - \frac{1}{6} \left[\nabla^2 n_\sigma - 2\tau_\sigma + \frac{1}{2} \frac{(\nabla n_\sigma)^2}{n_\sigma} \right] u^2 + \dots$$

Spherically
averaged

Laplacian

inter-particle
distance

kinetic energy density

$$\tau_\sigma = \sum_i |\nabla \varphi_{i\sigma}|^2$$

* following Becke's notation

Becke and Russel, PRA 39, 3761 (1989)

SR behavior of the c-hole: Becke, JCP 88, 1054 (1989)

Self-interaction free Hx

Exact short-range behavior of the x-hole

$$n_x^\sigma(\mathbf{r}, u) = -n_\sigma - \frac{1}{6} \left[\nabla^2 n_\sigma - 2\tau_\sigma + \frac{1}{2} \frac{(\nabla n_\sigma)^2}{n_\sigma} \right] u^2 + \dots$$

Self-Interaction-free Hx

$$n^\sigma(\mathbf{r}, u) + n_x^\sigma(\mathbf{r}, u) = 0$$

vanishes for one orbital

$$D_\sigma = \tau_\sigma - \frac{1}{4} \frac{(\nabla n_\sigma)^2}{n_\sigma}$$

vanishes for one orbital

Importance of fulfilling exact conditions (2/2)

PRL 115, 036402 (2015)

PHYSICAL REVIEW LETTERS

week ending
17 JULY 2015

Strongly Constrained and Appropriately Normed Semilocal Density Functional

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(Received 12 April 2015; published 14 July 2015)

The ground-state energy, electron density, and related properties of ordinary matter can be computed efficiently when the exchange-correlation energy as a functional of the density is approximated semilocally. We propose the first meta-generalized-gradient approximation (meta-GGA) that is fully constrained, obeying all 17 known exact constraints that a meta-GGA can. It is also exact or nearly exact for a set of “appropriate norms,” including rare-gas atoms and nonbonded interactions. This strongly constrained and appropriately normed meta-GGA achieves remarkable accuracy for systems where the exact exchange-correlation hole is localized near its electron, and especially for lattice constants and weak interactions.

Hierarchy of DFT approx.s



MBP-diagrams:

Hybrids:

Meta-GGAs:

GGAs:

LDAs:

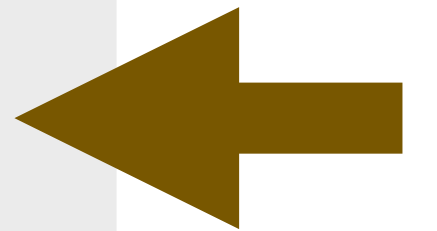
Unoccupied orbitals / eigenvalues

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$$E_{xc}[n] \approx \int d^3r \epsilon_{xc}(n(\mathbf{r}), \nabla n(\mathbf{r}), \tau(\mathbf{r}))$$

$$E_{xc}[n] \approx \int d^3r \epsilon_{xc}(n(\mathbf{r}), \nabla n(\mathbf{r}))$$

$$E_{xc}[n] \approx \int d^3r \epsilon_{xc}(n(\mathbf{r}))$$



$$E_x^{\text{Fock}} = - \sum_{\sigma} \sum_j \int \int d^2r d^3r' \frac{|\sum_i \varphi_{j\sigma}^*(\mathbf{r}) \varphi_{j\sigma}(\mathbf{r}')|^2}{2|\mathbf{r} - \mathbf{r}'|}$$

Density-functional thermochemistry. III. The role of exact exchange

Axel D. Becke

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(Received 30 October 1992; accepted 16 December 1992)

Despite the remarkable thermochemical accuracy of Kohn–Sham density-functional theories with gradient corrections for exchange-correlation [see, for example, A. D. Becke, *J. Chem. Phys.* **96**, 2155 (1992)], we believe that further improvements are unlikely unless *exact-exchange* information is considered. Arguments to support this view are presented, and a semiempirical exchange-correlation functional containing local-spin-density, gradient, and exact-exchange terms is tested on 56 atomization energies, 42 ionization potentials, 8 proton affinities, and 10 total atomic energies of first- and second-row systems. This functional performs significantly better than previous functionals with gradient corrections only, and fits experimental atomization energies with an impressively small average absolute deviation of 2.4 kcal/mol.

Aiming at better non-local xc-hole features

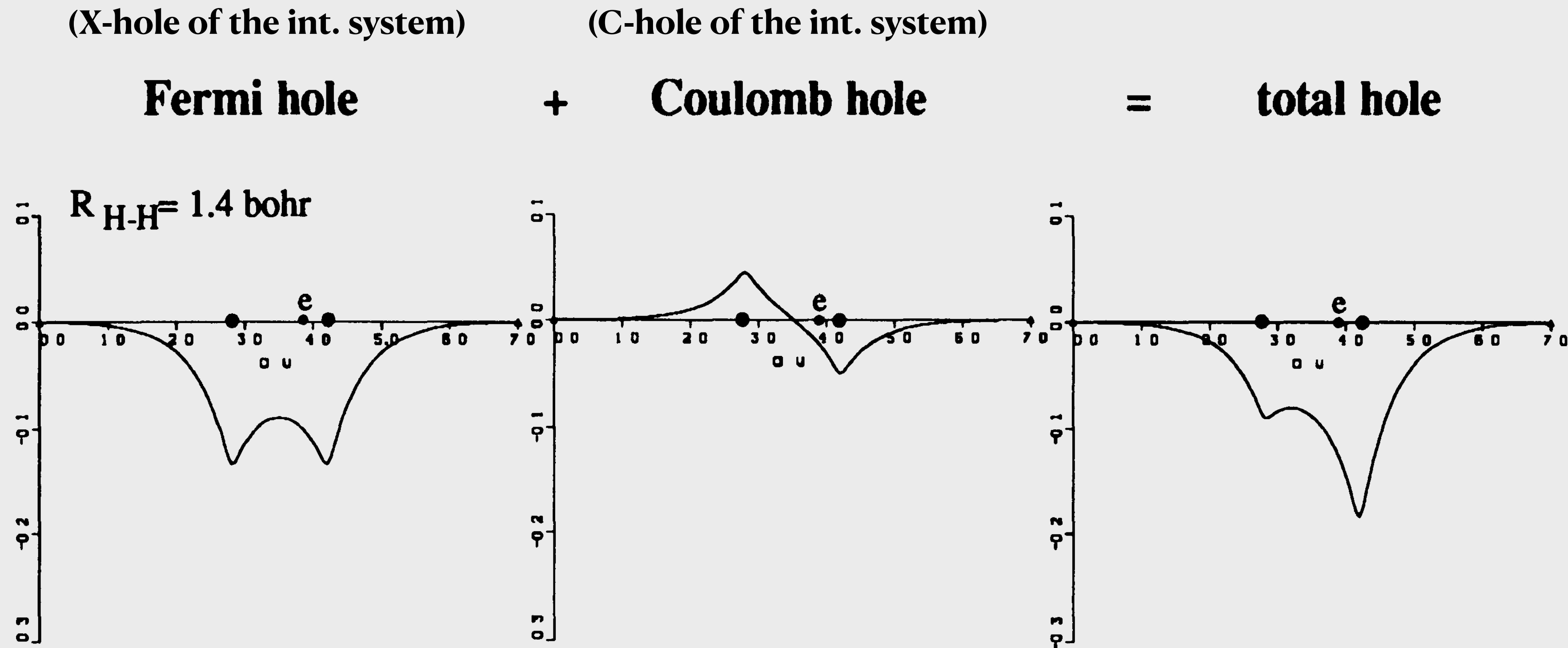


Fig 4 4 Fermi hole, Coulomb hole and total hole in the hydrogen molecule at various values of the internuclear distance In all plots the reference electron is placed 0.3 bohr at the left of the right H-atom

Figure from:

Buijse and Baerends D.E. Ellis (ed.), Density Functional Theory of Molecules, Clusters, and Solids, 1-46. 1995
Kluwer Academic Publishers

Popular hybrids

PBE0 (or PBEh)

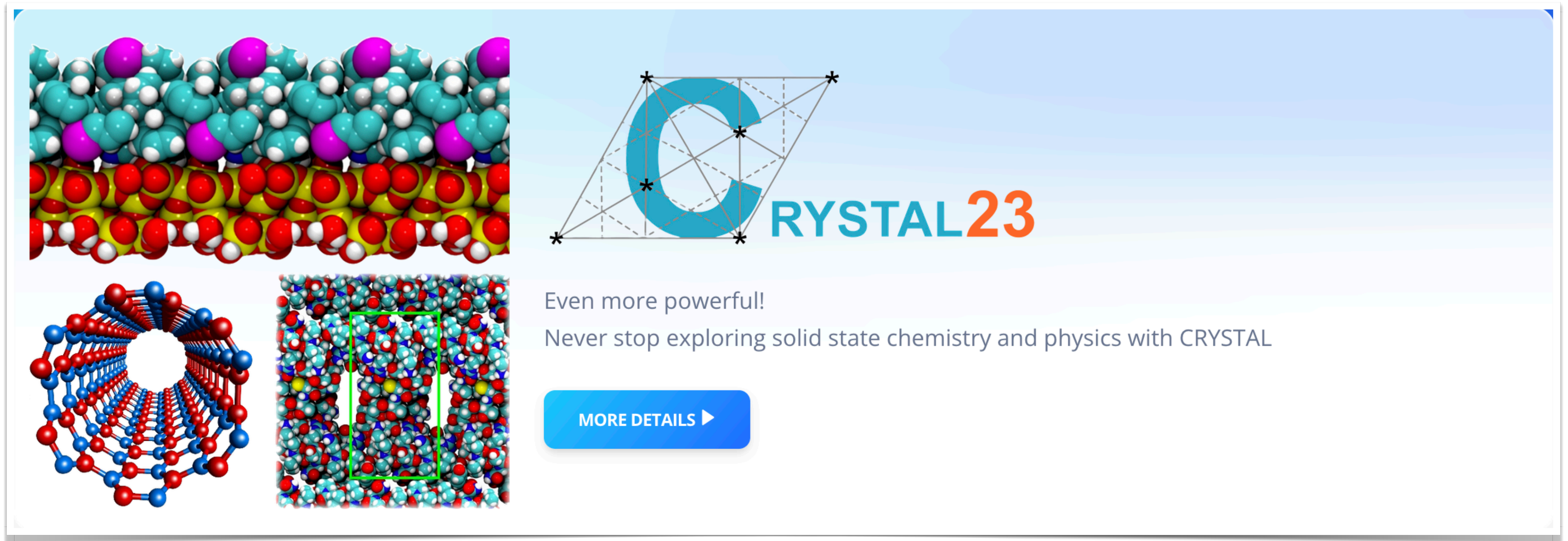
$$E_{\text{x}}^{\text{PBE0}} = \frac{1}{4}E_{\text{x}}^{\text{exact}} + \frac{3}{4}E_{\text{x}}^{\text{PBE}}$$
$$E_{\text{c}}^{\text{PBE0}} = E_{\text{c}}^{\text{PBE}}$$

B3LYP:

$$E_{\text{xc}}^{\text{B3LYP}} = a_0 E_{\text{x}}^{\text{exact}} + (1 - a_0) E_{\text{x}}^{\text{LSDA}} + a_{\text{x}} \Delta E_{\text{x}}^{\text{B88}} + (1 - a_{\text{c}}) E_{\text{c}}^{\text{VWN-RPA}} + a_{\text{c}} E_{\text{c}}^{\text{LYP}}$$

Have fun with DFT

Financial support for this lecture



The advertisement banner for CRYSTAL23 features a light blue background with a subtle gradient. On the left, there are three molecular models: a top layer of purple, cyan, and white spheres; a middle layer of red and yellow spheres; and a bottom layer of red and blue spheres. To the right of these models is a large, stylized blue 'C' logo with a white outline, and the text 'RYSTAL23' in blue and orange. Below the logo, the text 'Even more powerful!' and 'Never stop exploring solid state chemistry and physics with CRYSTAL' is displayed. A blue button with the text 'MORE DETAILS' and a right-pointing arrow is located at the bottom right of the banner.

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